{SUMMER INTERNSHIP PROJECT REPORT}

Deep Learning based Single Image Dehazing: A Comparative Evaluation of U-Net Architectures

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Abstract

Image dehazing is a fundamental preprocessing technique in computer vision that aims to restore clear, high-contrast images from scenes degraded by atmospheric scattering effects such as fog, haze, smoke, and mist. Traditional prior-based approaches exhibit limited performance in the presence of dense or non-uniform haze, often introducing colour distortions and halo artifacts because of their dependence on predefined statistical priors. To overcome these limitations, this work proposes and systematically investigates a set of deep learning-based architectures for single-image dehazing, marking a shift from handcrafted heuristics toward end-to-end data-driven feature learning. Specifically, three U-Net variants are investigated: a Baseline U-Net, a Channel Attention U-Net utilizing Squeeze-and-Excitation blocks, and a lightweight Depthwise Separable U-Net variant incorporating FFT-based frequency-domain refinement for enhanced structural recovery. Comprehensive experiments on the indoor ITS and outdoor RESIDE-6K benchmarks demonstrate that employing a Structural Similarity Index Measure (SSIM)-based loss function consistently improves dehazing performance compared with the conventional Mean Squared Error (MSE) loss. The top-performing Channel Attention U-Net achieved 28.54 dB PSNR and 0.9651 SSIM on the RESIDE-6K dataset. Furthermore, to address practical deployment constraints, the proposed Depthwise Separable architectures reduced the trainable parameter count by approximately 81% (from ~31 million to ~million parameters) while preserving highly competitive visual fidelity. These results demonstrate that lightweight architectures can achieve strong dehazing performance at a fraction of the computational cost, offering a scalable solution for downstream applications such as autonomous driving and outdoor surveillance.

1. Introduction

1.1 Introduction

Image dehazing is a low-level image restoration task in computer vision whose objective is to recover a clear, haze-free image from a single observation that has been visually degraded by atmospheric phenomena such as haze, fog, smoke, dust, or mist. As light travels from a scene point to the camera through such a medium, suspended particles scatter and absorb part of it, while ambient light from other directions is simultaneously scattered into the line of sight. The captured image is therefore not simply a darkened or blurred version of the true scene; it is a superposition of an attenuated scene signal and an additive, depth-dependent veiling component. Because this veiling component grows with distance, haze degradation is inherently non-uniform across a frame: pixels corresponding to distant objects are far more heavily affected than pixels belonging to nearby objects within the very same photograph.

Framed this way, dehazing is fundamentally a signal-recovery problem rather than a purely cosmetic one. The true scene radiance is present in the observed image, but it has been mixed with an additive airlight component and attenuated multiplicatively by the same scattering process; any effective dehazing method, whether built on handcrafted statistical priors or on learned neural representations, must therefore implicitly or explicitly disentangle these two competing signals before a faithful reconstruction becomes possible. Atmospheric degradation of this kind arises whenever light travels through a medium containing suspended particulate matter. Haze consists of fine dust, smoke, and other dry particles suspended in the air, while fog and mist are formed primarily by suspended water droplets; smoke, whether from combustion or industrial sources, contributes absorptive and scattering particles of its own. Although these phenomena differ in physical origin and particle composition, they share the same optical consequence described above, and the relative strength of the attenuation and airlight terms in any given pixel depends on the density of the atmospheric medium and the distance between the camera and the corresponding scene point.

The effect of this degradation on image quality is substantial and multi-dimensional. Contrast is reduced because the scattered airlight adds a roughly uniform veiling luminance across the frame, compressing the dynamic range of true scene radiance values. Colours appear washed out or shifted because the scattering coefficient of the atmosphere is wavelength-dependent, altering the relative intensities of the red, green, and blue channels in a way that is not present in the original scene. Edges and fine textures lose sharpness because the additive scattering component behaves like a low-pass, depth-dependent veil superimposed on the image content, and beyond a certain haze density, distant structures can become almost indistinguishable from the sky or background. Because so many downstream tasks, object detection, semantic segmentation, depth estimation, tracking, and scene recognition, depend on receiving inputs with reliable contrast, colour, and structural detail, haze degradation is not merely a nuisance to human viewers; it is a substantive obstacle to the reliable operation of vision systems in outdoor and uncontrolled environments.

Image restoration, of which dehazing is one instance, is therefore an essential preprocessing and enhancement discipline within computer vision. Rather than simply improving an image for human viewing, restoration techniques aim to recover an estimate of the underlying scene as faithfully as possible, so that both human observers and automated algorithms can operate on information closer to ground truth. Early approaches to dehazing were built around explicit physical models of atmospheric scattering combined with handcrafted statistical priors, assumptions about natural image statistics that allow the unknown parameters of the scattering model to be estimated from a single hazy image without any additional information. While these methods established the theoretical foundation of the field and remain instructive for understanding the physics of haze formation, they are fundamentally limited by the validity of their underlying assumptions, and they tend to fail in precisely the complex, real-world conditions where dehazing is most needed.

Over the past decade, the field has undergone a decisive shift from these handcrafted, model-driven techniques toward data-driven deep learning approaches. Rather than relying on a fixed statistical prior, convolutional neural networks and, more recently, attention-based and transformer-based architectures learn the mapping between hazy and haze-free images directly from large collections of paired training data. This shift has enabled dehazing systems to generalize far more effectively across diverse haze densities, lighting conditions, and scene compositions, and it has opened the door to architectures that jointly optimize for restoration accuracy, structural fidelity, and computational efficiency, the central themes explored throughout the remainder of this report.

1.2 Need for Image Dehazing

Every downstream vision task, detection, segmentation, tracking, depth estimation, implicitly assumes that its input faithfully represents scene structure. Haze violates that assumption nonuniformly across the frame, since attenuation grows with depth: a pedestrian standing close to a camera and a vehicle several hundred metres away can be degraded to very different extents within the very same image. This spatially non-uniform character of the problem is one of the central difficulties that any dehazing method must contend with, and it is a recurring theme across the application domains discussed below.

1.2.1 Autonomous Driving

Autonomous vehicles rely on camera-based perception pipelines to detect lanes, pedestrians, other vehicles, and traffic signage in real time. Haze and fog reduce the effective visual range of onboard cameras, lower the contrast between foreground objects and the background, and can cause object detectors to miss distant hazards or misclassify partially obscured objects. Because driving decisions must be made within tight latency budgets and with a very low tolerance for error, even a modest degradation in image quality can translate into a meaningful increase in perception risk. Effective dehazing restores the contrast and structural detail that downstream detection and tracking modules depend on, making it a critical component of robust perception stacks for advanced driver-assistance and autonomous driving systems, particularly in regions where fog and haze are common seasonal occurrences.

1.2.2 Intelligent Transportation Systems

Beyond individual vehicles, intelligent transportation systems rely on networks of roadside and traffic cameras to monitor congestion, detect incidents, enforce traffic regulations, and manage adaptive signal timing. These systems typically operate continuously and cannot be repositioned or paused when atmospheric conditions deteriorate. Haze or smog over a stretch of highway can significantly impair automatic number-plate recognition, vehicle counting, and incident detection algorithms that were trained or calibrated on clear-weather footage. Dehazing algorithms embedded within the video processing pipeline allow such systems to maintain a consistent level of accuracy regardless of ambient visibility, which is essential for the reliability of city-scale traffic management infrastructure.

1.2.3 Surveillance

Security and surveillance systems are expected to provide continuous, dependable monitoring of outdoor spaces such as public squares, transportation hubs, borders, and industrial facilities. Haze, fog, and smoke can severely limit the effective range at which a surveillance camera can resolve faces, licence plates, or suspicious activity, undermining the very purpose of the system precisely when visibility-related incidents, such as fires or accidents, are most likely to occur. Since surveillance footage is frequently used for later forensic analysis as well as real-time monitoring, restoring clarity through dehazing improves both the immediate situational awareness of operators and the long-term evidentiary value of recorded footage, particularly over the long distances typical of outdoor security deployments.

1.2.4 Robotics

Mobile robots operating outdoors, whether in agricultural fields, construction sites, or disaster response scenarios, depend on visual sensing for navigation, obstacle avoidance, and manipulation tasks. Environmental haze, dust clouds kicked up by machinery, or smoke from a fire scene can all degrade the visual input available to a robot's perception system in ways that are functionally similar to atmospheric haze. Because robots often must operate autonomously in precisely the hazardous or dusty conditions where visibility is compromised, robust dehazing is not an optional enhancement but a prerequisite for safe and reliable operation in these settings.

1.2.5 Drone Navigation

Unmanned aerial vehicles conducting aerial photography, inspection, or delivery missions frequently operate at altitudes and over distances where atmospheric haze has a pronounced effect on image quality, since the scattering-induced degradation intensifies with the distance between the camera and the imaged scene. A drone attempting to navigate using visual odometry, avoid obstacles, or identify a landing zone through hazy air is at increased risk of misjudging depth and distance. Dehazing algorithms that can operate efficiently onboard, given the limited computational and power budgets of aerial platforms, are therefore particularly valuable for extending the operational envelope of drones into hazier atmospheric conditions.

1.2.6 Remote Sensing

Satellite and high-altitude aerial imagery used for environmental monitoring, agricultural assessment, urban planning, and disaster response is frequently affected by atmospheric haze, thin cloud cover, and aerosol scattering over very long optical paths. Because remote sensing imagery is often analyzed automatically, for example, to classify land cover, estimate vegetation indices, or detect changes over time, haze-induced distortions in reflectance values can propagate directly into inaccurate downstream measurements. Dehazing serves as an important preprocessing step in remote sensing pipelines, improving the radiometric consistency of imagery captured under varying atmospheric conditions and across different acquisition dates.

1.2.7 Medical Imaging

Although atmospheric haze in the strict sense does not apply to internal medical imaging modalities, a closely related degradation appears in contexts such as endoscopic and laparoscopic surgical video, where lens fogging, surgical smoke, and tissue-scattering effects introduce a hazelike veiling and loss of contrast highly analogous to atmospheric scattering. Restoring clarity in such video is directly relevant to surgical safety, since surgeons rely on clear visual feedback to guide delicate procedures. Techniques and architectural principles developed for atmospheric image dehazing have, as a result, found meaningful application in enhancing visibility within these specialized medical imaging contexts.

1.2.8 Smart Cities

Smart city infrastructure integrates numerous outdoor camera networks for purposes ranging from environmental monitoring and air-quality estimation to public safety and infrastructure management. Given that urban haze and smog are themselves indicators of air pollution levels, dehazing serves a dual purpose in this context: it both restores usable imagery for other computer vision applications operating on the same camera feeds, and, in some research systems, the estimated haze density or transmission map is itself used as a proxy signal for local air-quality estimation. As cities increasingly depend on distributed visual sensing for automated decisionmaking, the ability to maintain image quality under variable atmospheric conditions becomes a foundational requirement of the broader smart city sensing ecosystem.

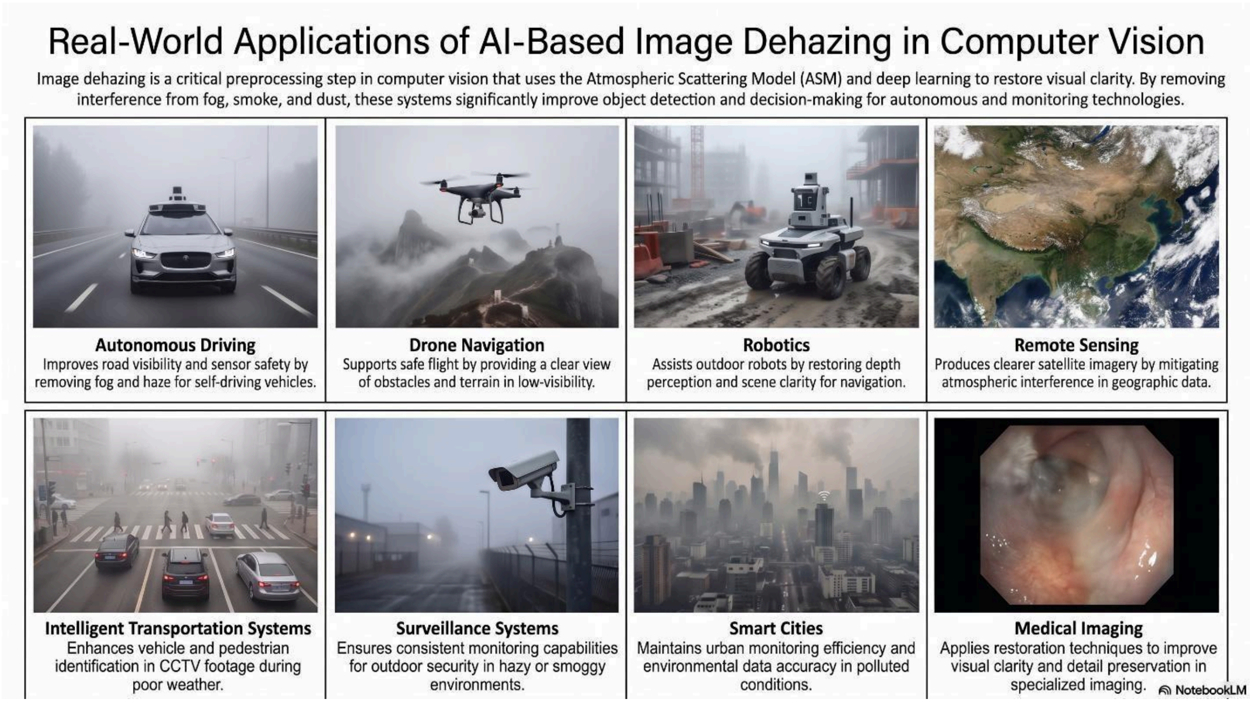


Figure 1.1: Application domains that depend on effective image dehazing

1.3 Atmospheric Scattering Model

Nearly every physics-grounded dehazing method, classical or deep-learning-based, is built on the atmospheric scattering model (ASM), which mathematically explains how a hazy image is formed. Haze is formed when the atmosphere contains a sufficiently high concentration of small particles, water droplets, dust, smoke, or aerosols, whose physical dimensions are comparable to the wavelength of visible light. As light travels from a scene point toward the camera, it interacts with these particles through the physical process of scattering, in which a fraction of the light's energy is redirected away from its original path. Some of this scattered light is lost from the line of sight entirely, attenuating the light that originated at the scene point, while ambient light from other directions, principally sunlight and skylight, is simultaneously scattered into the line of sight, adding a veiling component that did not originate at the scene point at all. This latter contribution is commonly referred to as airlight, and it is the dominant reason hazy images appear washed out and low in contrast, since the airlight component tends toward the colour of the ambient atmospheric illumination rather than the true colour of the scene.

The relationship between these physical quantities was first formalized for horizontal visibility by Koschmieder, and later adapted into the imaging formulation used throughout modern dehazing research, which expresses the observed hazy image intensity I(x) at pixel location x as a combination of the attenuated scene radiance and the airlight contribution, an equation commonly referred to as Koschmieder's law:

$$I(x) = J(x)t(x) + A(1 - t(x))$$

Here, J(x) denotes the haze-free scene radiance that would have been recorded in the absence of atmospheric interference, the quantity a dehazing algorithm ultimately seeks to recover. The first term, J(x)t(x), is the direct attenuation component: the true scene signal weakened by scattering. The second term, A(1 − t(x)), is the airlight component contributed by ambient scattered light, and it grows as transmission decreases. Recovering the clear image J(x) therefore reduces to the accurate estimation of t(x) and A, which is precisely the estimation problem that both classical priors and modern learning-based dehazing networks are designed to solve. The transmission map itself is related to scene depth through Beer–Lambert-type exponential attenuation:

$$t(x) = e^{-\beta d(x)}$$

In this expression, β is the atmospheric scattering coefficient, which quantifies the optical density of the haze medium, larger values of β correspond to thicker, more strongly scattering haze, and d(x) is the scene depth at pixel x, measured as the distance between the camera and the corresponding scene point. Table 1.1 summarizes each symbol used in the model for quick reference.

Symbol	**Meaning**
I(x)	Observed hazy image intensity at pixel location x.
J(x)	True haze-free scene radiance at pixel x, the quantity a dehazing algorithm seeks to recover.
A	Global atmospheric light, generally assumed constant across a single image.
t(x)	Transmission map, the fraction of light originating at x that reaches the camera without being scattered, $t(x) \in [0, 1]$.

β	Atmospheric scattering coefficient, characterizing the optical density of the haze medium.
d(x)	Scene depth at pixel x, the distance between the camera and the corresponding scene point.

Table 1.1: Definition of symbols used in the atmospheric scattering model

This decomposition explains an important qualitative property of haze: because t(x) decays exponentially with depth, distant scene content becomes dominated almost entirely by the airlight term A(1 − t(x)), which carries no information about the original scene. This is why dense, farfield haze is fundamentally harder to reverse than thin, near-field haze, the signal-to-noise ratio between scene radiance and airlight collapses as depth grows, and nearby, shallow-depth regions retain high transmission and closely resemble the true scene, while distant, deep regions are dominated by airlight and appear washed out. Together, these two equations constitute the classical atmospheric scattering model that underlies both traditional prior-based dehazing methods, which attempt to estimate A and t(x) explicitly, and many deep learning architectures, which either learn to invert this model directly or use it implicitly as a physical prior to guide network design and loss formulation.

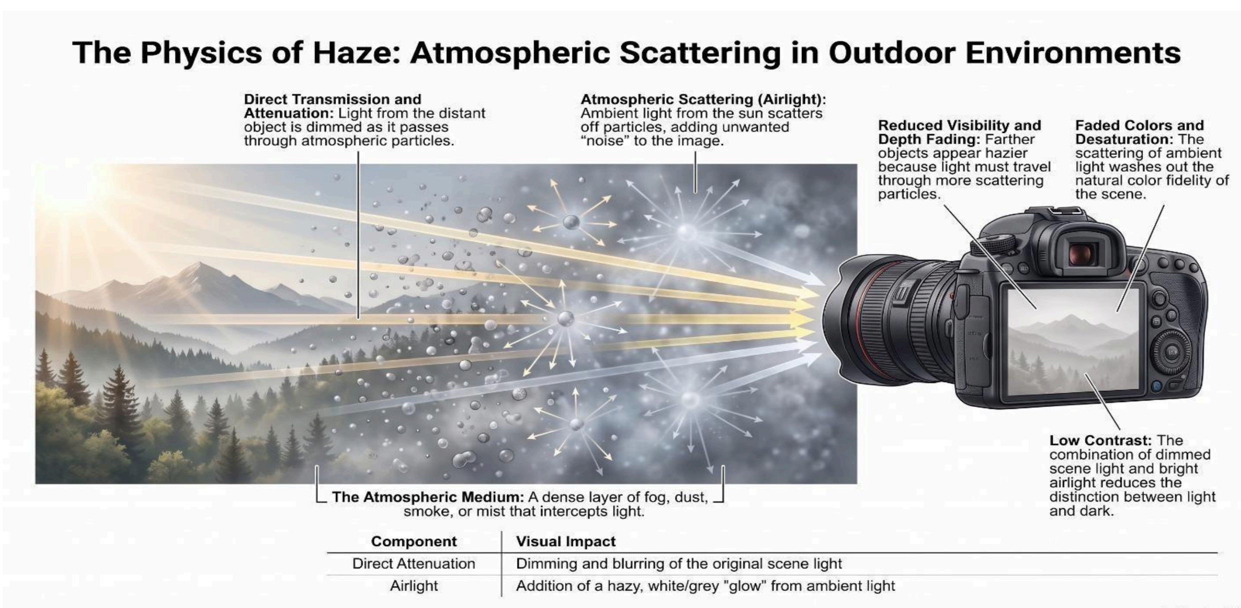


Figure 1.2: Atmospheric Scattering Model

1.4 Challenges in Image Dehazing

Despite decades of research, image dehazing remains a technically challenging problem because the atmospheric scattering model described in Section 1.3 is fundamentally under-constrained: given only a single hazy observation I(x), both the scene radiance J(x) and the transmission map t(x) are unknown, meaning that recovering a clear image requires additional assumptions, priors, or learned knowledge to resolve this ambiguity. Several specific characteristics of real-world haze compound this fundamental difficulty.

Dense haze presents one of the most severe challenges, since as haze density increases, the transmission t(x) approaches zero across large portions of the image, causing the observed intensity to be dominated almost entirely by the airlight term and leaving very little residual information about the true scene content to recover. Beyond a certain density threshold, this loss of information becomes effectively irreversible using single-image cues alone, and even state-of-the-art learning-based methods can only plausibly infer detail rather than truly recover it.

Non-uniform haze, in which fog or smoke density varies unevenly across the scene due to wind, localized sources of smoke, or patchy fog formation, violates the common simplifying assumption of a spatially smooth, depth-dependent transmission map, making it substantially harder for both handcrafted priors and learned models to correctly separate the true scene signal from the haze component. Low visibility conditions more broadly reduce the amount of usable structural and textural information present in the input image, which is problematic because most dehazing algorithms rely on some form of local contrast or edge information to guide the restoration process.

Colour distortion arises because the scattering coefficient of atmospheric particles is wavelength dependent, meaning that different colour channels are attenuated to different degrees; algorithms that fail to model this wavelength dependence accurately can introduce unnatural colour casts into the restored image. Halo artifacts, visible ringing or brightness discontinuities around high-contrast edges, particularly at object boundaries with abrupt depth changes, are a well-documented failure mode of both patch-based traditional methods and, to a lesser extent, poorly regularized neural networks, and they arise from incorrect transmission estimates near depth discontinuities. Closely related is the problem of edge loss, in which fine structural detail is smoothed away either by the haze itself or by overly aggressive restoration and denoising operations applied during dehazing.

Complex illumination conditions, including non-uniform lighting, strong shadows, and mixed natural and artificial light sources, further complicate the separation of true scene radiance from atmospheric scattering effects, since many dehazing assumptions implicitly presume relatively uniform ambient illumination. Finally, computational challenges are significant for real-time and embedded applications: while some traditional methods are lightweight, more accurate patchbased priors are often too slow for live video, and while learning-based methods can achieve strong accuracy, achieving that accuracy with a network small and fast enough for deployment on resource-constrained platforms such as vehicles or drones remains an active area of research and a central motivation for the lightweight architectural variant proposed in this project.

1.5 Traditional Image Dehazing Methods

Before the widespread adoption of deep learning, image dehazing research was dominated by methods that either enhanced image contrast directly, without reference to any physical model of haze formation, or that explicitly estimated the parameters of the atmospheric scattering model using handcrafted statistical priors derived from empirical observations about natural images. Understanding these traditional approaches is important not only for historical context but also because several of their underlying principles, particularly the atmospheric scattering formulation itself, continue to inform the design and evaluation of modern deep learning architectures. Table 1.2 summarizes four representative traditional techniques, along with their working principles, advantages, and limitations.

Method	Working Principle	Advantages	Limitations
Histogram Equalization	Redistributes image intensity levels so the output histogram is approximately uniform, stretching dynamic range and improving global contrast.	Computationally inexpensive; simple to implement; improves overall visibility in mildly hazy images.	Operates only on intensity/contrast statistics and does not model haze formation; tends to overenhance noise and cause colour shifts in dense haze.
Method	**Working Principle**	**Advantages**	**Limitations**

Contrast Stretching	Linearly maps a narrow range of pixel intensities to the full available dynamic range to enhance perceptual contrast.	Simple and fast; requires no physical haze model or camera parameters.	Ignores spatially varying haze density and scene depth, often underrestoring distant regions while over-stretching nearby clear ones.
Dark Channel Prior (DCP)	Exploits the observation that, in most haze-free outdoor patches, at least one colour channel has very low intensity; this statistic is used within the atmospheric scattering model to estimate t(x) and A.	Physically grounded; produces visually convincing results on a wide range of natural outdoor images without any training data.	Fails in bright or sky regions where the darkchannel assumption is violated; computationally costly due to soft-matting refinement; struggles under dense, non-uniform haze.
Color Attenuation Prior (CAP)	Models a linear relationship between scene depth and the difference between an image's brightness and saturation, and learns the coefficients of this relationship to recover depth and transmission.	More computationally efficient than DCP-based soft matting; offers a straightforward, learnable depth-estimation strategy.	Accuracy depends on the validity of the assumed linear relationship, which does not hold uniformly in complex scenes with varied lighting and texture.

Table 1.2: Comparison of traditional image dehazing methods

As Table 1.2 illustrates, contrast-based methods such as histogram equalization and contrast stretching are computationally simple and require no physical modelling, but they operate purely on the statistics of pixel intensities and therefore cannot distinguish genuine scene structure from haze-induced degradation, often producing unnatural results under non-trivial haze conditions. Prior-based methods such as the Dark Channel Prior and Color Attenuation Prior instead work within the atmospheric scattering framework and generally produce more physically plausible restorations, but they depend on statistical assumptions that do not hold uniformly across all scene types, most notably in bright regions such as sky, snow, or light-coloured surfaces, where their core assumptions systematically break down.

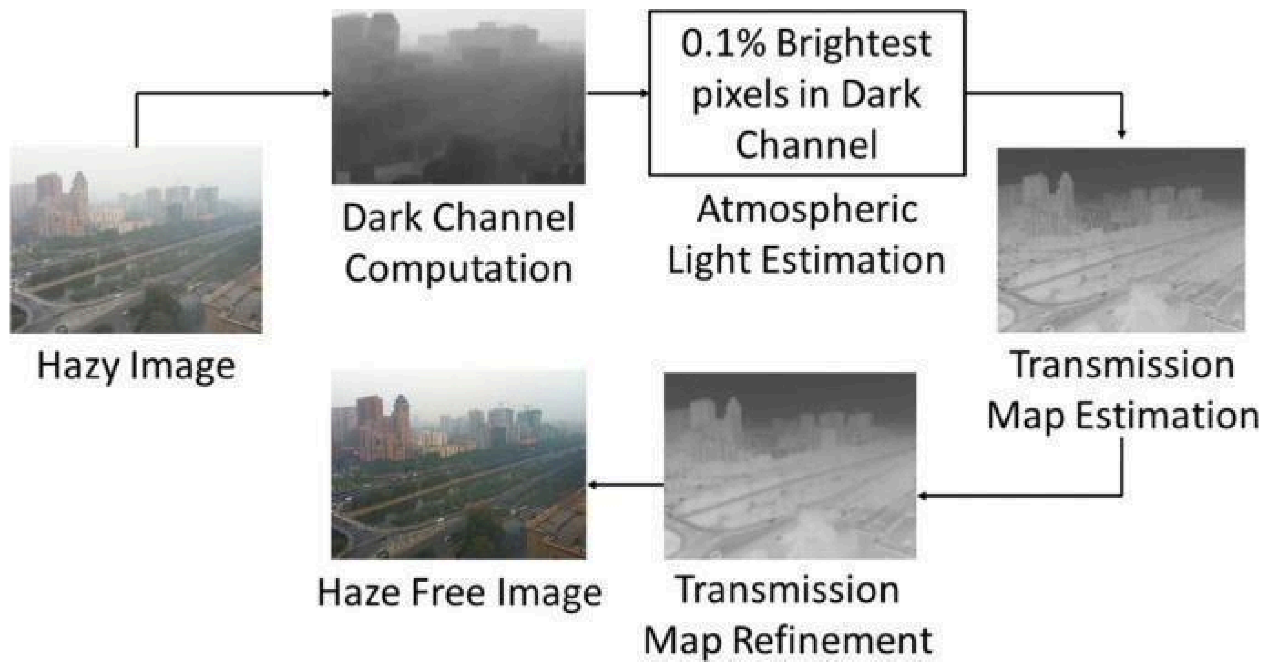


Figure 1.3: Illustrative Dark Channel Prior pipeline

The limitations of prior-based methods motivated the first generation of learning-based dehazing systems. Rather than estimating A and t(x) through handcrafted statistics, early convolutional networks such as DehazeNet and AOD-Net learned the transmission estimation step, and, eventually, the full ASM inversion, directly from paired hazy/clear training data. Subsequent architectures, including GridDehazeNet and FFA-Net, extended this idea with multi-scale feature fusion and feature-level attention to better capture haze that varies in density and colour across a scene. This progression, from fixed statistical assumptions toward representations learned end-to-end, sets the stage for the research gap discussed next and is examined in further depth in the literature survey that follows this chapter.

1.6 Research Gap

The limitations of traditional dehazing methods, examined in Section 1.5, motivate a broader research gap that this project, along with the wider field of deep learning-based dehazing, seeks to address. Handcrafted priors such as the Dark Channel Prior fundamentally rely on statistical regularities that are observed to hold across many natural outdoor images on average, but that are not guaranteed to hold for any specific image or scene region. When these assumptions fail, as they systematically do in sky regions, snow-covered scenes, brightly lit indoor environments, or scenes containing large uniformly coloured surfaces, the resulting transmission and atmospheric light estimates become inaccurate, producing visible artifacts, colour casts, or incomplete haze removal rather than a graceful degradation in quality.

Dense haze remains a persistent problem even for physically grounded methods, since as transmission approaches zero, the inverse problem of recovering scene radiance becomes numerically unstable regardless of how accurately the transmission map itself is estimated; small errors in the estimated transmission are amplified into large errors in the recovered radiance. Computational complexity is another significant limitation, since accurate implementations of prior-based methods typically require an additional refinement step, such as soft matting or guided filtering, which is often the most expensive part of the pipeline and scales poorly with image resolution. More fundamentally, traditional methods exhibit a pronounced lack of real-world generalization: because their assumptions are derived from statistical observations of natural outdoor imagery, they transfer poorly to atypical conditions such as underwater imagery, nighttime haze illuminated by artificial light, or synthetic and industrial environments not represented in the empirical studies underlying the original priors.

Early convolutional neural network approaches narrowed this gap considerably by learning the mapping between hazy and clear images directly from data, but they introduced limitations of their own, and a specific, architecturally precise gap remains even in standard encoder-decoder networks such as the U-Net. Standard U-Net-style architectures treat every feature channel with equal importance during reconstruction, giving the network no explicit mechanism to prioritize the channels that carry haze-relevant information over those carrying ordinary scene structure. Repeated downsampling in such spatial encoders progressively discards high-frequency detail, edges, textures, fine boundaries, that skip connections alone cannot fully restore, leaving outputs visibly blurred in some cases. This limitation is especially pronounced outdoors, where haze density and the resulting colour distortion vary channel-by-channel and region-by-region across the image, and a baseline convolutional block has no explicit way to determine which channels most need recalibration for a given input. Taken together, these observations point to the same underlying deficiency: baseline U-Net architectures lack any mechanism for feature-level, contentadaptive recalibration, and a convolution produces a fixed set of output channels regardless of whether haze is thin or dense, uniform or spatially varying. This is the specific gap that the proposed method, introduced in Section 3.8, is designed to close, while the broader need for strong generalization, robustness to dense and non-uniform haze, and architectural efficiency continues to motivate the lightweight variants explored later in this project.

1.7 Literature Review

1.7.1 Neural Augmentation Methods

Neural augmentation refers to the enhancement of model performance through the integration of additional neural network modules.

- Ju et al (2024)[25]- All-inclusive image enhancement (AIIE) unifies enhancement of degraded images affected by low-frequency corruption such as haze, low-light, underwater blur, and sandstorms. Using a statistical property of the discrete cosine transform (DCT) of 1,000 high- and 1000 low-quality images on their DCT domains. Which shows that the normalized DCT coefficients (between 0 and 1) of high-quality images has about 95% fall in the interval [0, 0.2]; for low-quality images, almost all the coefficients are in the same interval. This fundamental property, called the DCT prior (DCT-P), is used in the AIIE algorithm followed by the enhancement module analyzing the DCT domain, identifying deviations caused by haze, low-light, underwater blur, or sandstorms, and applying corrective transformations to restore visibility and contrast. This unified design allows the module to enhance multiple types of degraded images efficiently, with low computational cost and improved perceptual quality. Dehazing (I-HAZE, O-HAZE, D-HAZE) AIIE ranked top 3 in PSNR/SSIM, achieved second-best FADE, and had the fastest runtime (0.077s vs >0.1s for others).

- Asha et al. (2024)[29]- ODD-Net is a hybrid deep learning architecture for single image dehazing that decomposes the problem into physically meaningful sub-networks. The architecture introduces an Atmospheric Light Net (A-Net) to estimate atmospheric light using dilated convolution, batch normalization, and ReLU activation functions, and a T-Net to measure information transmission from objects to the camera using multiscale convolutions and nonlinear regression to construct a transmission map. The Deep Retinex component within the architecture further comprises a Residual Illumination Map Estimation Network (RIMEN) for multi-scale illumination estimation and a Channel and Spatial Attention Dehazing U-Net (CASDUN) for haze removal. By integrating the outputs from A-Net and T-Net through the atmospheric scattering model, ODD-Net merges physics-based sub-networks with deep learning refinement to improve generalization on both synthetic and real-world hazy scenes. ODD-Net addresses the limitations of existing techniques by combining physics-grounded estimation with data-driven refinement, demonstrating improved fidelity and visual quality compared to prior spatial, transform, and learning-based baselines on full-reference and no-reference evaluation metrics.

- Wang et al. (2024)[31]- UCL-Dehaze proposes real-world image dehazing via unsupervised contrastive learning, published in IEEE Transactions on Image Processing. UCL-Dehaze leverages unpaired real-world hazy and clear images within an unsupervised contrastive learning framework, eliminating the dependency on large-scale annotated paired training data. The method introduces contrastive regularization terms that pull dehazed representations toward clear image distributions while pushing them away from hazy image distributions, effectively guiding the model to learn domain-invariant features. By training on real-world unpaired data without ground-truth supervision, UCL-Dehaze achieves strong generalization to diverse real-world hazy conditions and outperforms prior supervised and semi-supervised dehazing approaches on both quantitative and perceptual evaluation metrics.

- Zhou el al (2023)[24]- An effective physical-prior-guided single image dehazing network via unpaired contrastive learning (PDUNet). The learning process of PDUNet consists of a pre-training stage on synthetic data (RESIDE) and fine-tuning stage on real (O-HAZE/I-HAZE and outdoor scenes). Mixed-prior modules, controllable zero-convolution modules, and unpaired contrastive regularization with hybrid transmission maps are used to fully utilize complementary advantages of both prior-based and learning-based strategies. Through leveraging physical prior statistics and vast unlabeled real-data, PDUNet exhibits excellent generalization and adaptability in handling real-world hazy scenarios. Extensive experiments on public dataset have demonstrated that the proposed method improves PSNR, NIQE, and BRISQUE values by an average of 0.33, 0.69 and 2.3, respectively, with comparable model efficiency compared to SOTA.

- Shyam & Yoo (2023)[28], A data efficient single image dehazing method via adversarial auto-augmentation and an extended atmospheric scattering model, presented at the IEEE/CVF International Conference on Computer Vision (ICCV 2023). This approach addresses the critical challenge of limited training data in dehazing by introducing an adversarial auto-augmentation strategy that synthetically generates diverse hazy training samples guided by an extended atmospheric scattering model. Rather than relying solely on large-scale paired datasets, the augmentation module learns to produce difficult yet realistic haze patterns during training, forcing the dehazing network to become more robust and data efficient. This neural augmentation of training data allows the model to achieve competitive dehazing performance with significantly fewer paired samples than conventional supervised methods, improving generalization to unseen real-world hazy conditions.

- Zheng et al. (2023)[30], Curricular Contrastive Regularization for Physics-Aware Single Image Dehazing (C2PNet), presented at the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2023). C2PNet integrates physical prior knowledge from the atmospheric scattering model with a contrastive learning regularization strategy that uses a curricular training schedule to progressively sample harder negative pairs. By grounding the contrastive learning objective in physical haze formation principles, C2PNet encourages the dehazing network to learn representations that are both perceptually meaningful and physically consistent. Training is conducted on RESIDE synthetic datasets with evaluation on SOTS indoor/outdoor splits and real-world hazy images including RTTS, demonstrating strong PSNR/SSIM improvements and superior perceptual quality compared to state-of-the-art supervised and contrastive dehazing baselines.

- Li et al (2022)[22]- A Dual-scale single image dehazing via neural augmentation where the transmission map and atmospheric light are estimated using model-based methods, followed by refinement through a dual-scale generative adversarial network (GAN). This model is trained using RESIDE synthetic haze datasets (ITS, OTS). Augmented with real-world hazy images (O-HAZE + ground truth pairs + including outdoor urban scenes) to improve generalization and the dual-scale design This model outperformed most CNN and GAN baselines, with Evaluation Metrics of DHQI: 62.58 (higher = better perceptual quality FADE: 0.5883 (lower = less fog density perceived).

- Li et al. (2022b)[27], Single image dehazing via model-based deep-learning, presented at the 2022 IEEE International Conference on Image Processing (ICIP). This work introduces a novel single image dehazing algorithm by integrating model-based and data-driven approaches. Model-based single image dehazing algorithms restore images with sharp edges and rich details at the expense of low PSNR values, while data-driven ones restore images with high PSNR values but with low contrast and even some remaining haze. Both transmission map and atmospheric light are initialized by the model-based methods and refined by deep learning-based approaches which form a neural augmentation, and haze-free images are then restored by using these estimated parameters. Experimental results validate that the proposed algorithm removes haze well from synthetic and real-world hazy images, and the proposed neural augmentation reduces the number of training data significantly. By anchoring the deep network's predictions in physically interpretable ASM parameters, the model achieves stable performance with smaller training sets compared to fully end-to-end methods, making it practical for scenarios where large-scale paired datasets are unavailable.

- Zhao et al (2021)[23]- A weakly supervised refinement framework for single image dehazing (RefineDNet) merge the merits of prior-based and learning-based approaches by dividing the dehazing task into two sub-tasks, i.e. visibility restoration and realness improvement. Firstly, it performs visibility restoration using a prior (the Dark Channel Prior) to estimate transmission and atmospheric light and produce a stable, coarse dehazed result and secondly, it applies a weakly supervised refinement stage an encoder–decoder refinement network trained adversarially with unpaired real hazy and clear images. RefineDNet with the perceptual fusion has an outstanding haze removal capability and can also produce visually pleasing results. Even implemented with basic backbone networks, RefineDNet can outperform supervised dehazing approaches as well as other state-of-the-art methods on indoor and outdoor datasets.

- Ju et al. (2021)[26]- Image dehazing and exposure (IDE) introduces a light absorption coefficient, into ASM, an enhanced ASM enhanced atmospheric scattering model (EASM which can address the dim effect and better model outdoor hazy scenes, and the gray world-assumption-based technique called IDE, IDE is based on a global stretch strategy, and it can simultaneously dehaze and expose single hazy images without the needs of post-processing. is integrated to enhance the visibility of hazy images, by recovering both the transmission map (for haze removal) and exposure parameters (for brightness correction) and trained using Synthetic paired datasets (RESIDE) plus real-world hazy/exposure-distorted images for validation. IDE achieves the highest R scores for almost all images, which is mainly due to the exposure ability inherited from EASM. Moreover, IDE also has the best score for NIQE; the recovery quality of IDE is remarkably better than that of other algorithms despite the lower PNSR score. EASM has a better modelling ability for hazy images compared to ASM, and the resultant IDE is superior to most state-of-the-art techniques in terms of both processing efficiency and recovery quality.

1.7.2 Attention-Based Dehazing Methods

Attention mechanisms can be used to enhance feature learning in deep neural networks for image dehazing. Attention allows models to focus on the most salient parts of an image by selectively emphasizing informative features and suppressing irrelevant ones. This capability is crucial for handling the complex degradation characteristics of hazy images, where visibility can vary significantly across different regions.

- ICAFormer (2025)[41], An Image Dehazing Transformer Based on Interactive Channel Attention, published in Sensors. To address the limitations of traditional dehazing algorithms in global feature association and local detail preservation, a novel Transformer-based dehazing model enhanced by an interactive channel attention mechanism is proposed. The architecture adopts a U-shaped encoder–decoder framework incorporating a feature extraction module and a feature fusion module based on interactive attention, where the interactive channel attention mechanism facilitates cross-layer feature interaction enabling the dynamic fusion of global contextual information and local texture details. The network leverages a multi-scale feature pyramid to extract image information across different dimensions, while an improved cross-channel attention weighting mechanism enhances feature representation in regions with varying haze densities. Extensive experiments on both synthetic and real-world datasets including the RESIDE benchmark demonstrate the superior performance of the proposed method, achieving PSNR gains of 0.53 dB for indoor scenes and 1.64 dB for outdoor scenes, alongside SSIM improvements of 1.4% and 1.7% respectively, compared with the second-best performing method.

- Zhang et al. (2025)[35]- A dual multi-scale network (DMSDN) to learn the dehazing knowledge from synthetical data designed to capture a large variety of objects, and then fine multi-scale blocks are designed to capture a small variety of objects at each scale. This Dual multiscale architecture processes feature several spatial scales in parallel and fuses them through complementary paths so the network can model both coarse atmospheric light and fine scene details. The multiscale modules apply convolutions or feature extractors at different spatial scales (e.g., small and large kernels or down/up sampling branches) and Multi-scale fusion aggregates features across resolutions to reconstruct details while keeping global consistency. The model fuses dual-path outputs via learned fusion layers and uses skip connections (U-Net style) to preserve low-level cues and stabilize gradient flow during training. Using RESIDE-Indoor (ITS) and RESIDE-Outdoor (OTS) for training and SOTS-Indoor/Outdoor (500 images each) for testing. DMSDN has higher PSNR and SSIM values indicate better fidelity and structural similarity also it Outperformed all baselines models (EPDN, FAMED-Net, GDN, AIRNET, MSBDN, BPPNET, and DCP) on SOTS datasets, showing clearer edges, more natural colors, and stronger quantitative. Note: the reference list dates this publication to IEEE Access 11, 2023.

- Lu et al. (2024)[33]- MixDehazeNet introduces a compact dehazing architecture with mix structural design that fuses multi-scale parallel large kernels with an enhanced parallel attention mechanism, the block runs several large kernels at different scales in parallel and attention branches run in parallel (not serially) so the model can preserve partial textures and fine details and also adapt to uneven haze distributions across the image. The enhanced parallel attention efficiently deals with uneven hazy distribution and allows useful features to pass through the backbone. Standard dehazing benchmarks including SOTS indoor/outdoor and other common datasets referenced in dehazing literature. substantial found PSNR/SSIM improvements, with the SOTS indoor PSNR reported around 42.62 dB, outperforming prior state-of-the-art in their comparison becoming the first method to exceed 42dB PSNR in the RESIDE-IN datasets.

- Hua et al. (2024)[34]- An end-to-end Lightweight Dual Attention Network (LWDA-Net) for direct recovery of haze-free images. The LWDA-Net architecture has the novel Dual Attention (DA) module, which takes into account completely different weighting information between samples and provides stronger regularization. The DA module is inserted early to modulate features before deeper processing. The DA block uses different kernel sizes and unequal weighting to provide stronger regularization and flexible receptive fields, with retaining shallow features and applying attention that can change local receptive fields, LWDA-Net aims to better preserve fine details (edges, textures) while removing global haze effects. Evaluation on standard dehazing datasets, reported that LWDA-Net outperforms several prior methods both quantitatively (PSNR/SSIM) and qualitatively, with specific mention of gains on SOTS. emphasis on a favorable tradeoff between restoration quality and model efficiency (parameter count and inference speed), making it suitable for embedded or real-time scenarios.

- Chen et al. (2024)[40]— DEA-Net addresses single image dehazing based on Detail-Enhanced Convolution and Content-Guided Attention, published in IEEE Transactions on Image Processing. The Detail-Enhanced Attention Block (DEAB) combines Detail-Enhanced Convolution (DEConv), which integrates difference convolutions to complement standard convolutions and enhance representation capacity, with a Content-Guided Attention (CGA) module that guides feature aggregation based on haze-relevant content, enabling the network to selectively emphasise detail-rich regions. Through re-parameterization, DEConv is equivalently converted into a vanilla convolution at inference to reduce parameters and computational cost, making the model efficient at deployment without sacrificing restoration quality. DEA-Net is evaluated on standard benchmarks including SOTS indoor/outdoor and HAZE4K, achieving strong PSNR and SSIM scores and demonstrating that structural innovation within convolution and attention design can yield significant gains without increasing model complexity at inference time.

- Song et al. (2023)[37], DehazeFormer proposes Vision Transformers for Single Image Dehazing, published in IEEE Transactions on Image Processing. While CNN-based methods have dominated image dehazing, vision Transformers had not yet been systematically explored for this task, and starting with the popular Swin Transformer, several of its key designs were found to be unsuitable for image dehazing. To this end, DehazeFormer consists of various improvements including a modified normalization layer, activation function, and spatial information aggregation scheme, and multiple variants are trained on various datasets to demonstrate effectiveness. The large DehazeFormer model was the first Transformer-based method to achieve PSNR over 40 dB on the SOTS indoor set, dramatically outperforming previous CNN-based state-of-the-art methods, and a large-scale realistic remote sensing dehazing dataset was also collected for evaluating the method's capability to remove highly non-homogeneous haze.

- Cui at al. (2023)[32]- ECANet is an enhanced context aggregation network for single image dehazing. Based on encoder–decoder structure, the ECANet improves feature representation by three feature aggregation blocks (FABs) on each scale. The FAB is a new efficient feature extraction module, which adequately extracts content features and style features due to the difference in receptive field between dilated convolution and ordinary convolution. And these complementary features are combined using spatial and channel attention mechanisms to each FAB. After the decoding process, an enhancing block to further refine image details under the supervision of clear references is done. ECANet benchmarks on both synthetic (RESIDE/SOTS, D-Hazy) and real datasets (RTTS, RW-Haze, O-HAZE/I-HAZE) to demonstrate both numerical gains (PSNR/SSIM) and perceptual improvements. Dilated + Standard convolution captures both global haze patterns and local edges. Spatial + Channel attention inside FABs yields focused correction of hazy regions with modest computational overhead. Lightweight design provides efficiency while improving perceptual quality.

- Dong et al. (2023)[39], Multi-Scale Attention Feature Enhancement Network (MSAFEN) is an end-to-end image dehazing network based on multi-scale feature enhancement, proposed to solve the problem of color distortion and loss of detail information common in most dehazing algorithms. Firstly, a feature extraction enhancement module captures the detailed information of hazy images and expands the receptive field; secondly, the channel attention mechanism and pixel attention mechanism of the feature fusion enhancement module dynamically adjust the weights of different channels and pixels; thirdly, a context enhancement module enhances context semantic information, suppresses redundant information, and obtains the haze density image with higher detail. MSAFEN is evaluated on SOTS indoor/outdoor splits and real-world hazy images, achieving a PSNR of 33.74 dB, SSIM of 0.9843, and LPIPS distance of 0.0040 on the SOTS-outdoor dataset, demonstrating better dehazing performance while preserving more image details and achieving color fidelity compared to prior multi-scale approaches.

- Qin et al. (2020)[36], FFA-Net is an end-to-end feature fusion attention network for single image dehazing that introduces a novel Feature Attention (FA) module combining Channel Attention with Pixel Attention. The FA module treats different features and pixels unequally, providing additional flexibility in handling varying types of information across the image and expanding the representational capacity of the network. A basic block structure combines Local Residual Learning and Feature Attention, where Local Residual Learning allows less important information such as thin haze regions or low-frequency content to bypass through multiple local residual connections, letting the main network focus on more effective information. An Attention-based different levels Feature Fusion (FFA) structure adaptively learns feature weights from the FA module, giving more weight to important features and retaining the information of shallow layers to pass into deep layers. On the SOTS indoor benchmark, FFA-Net boosted the best published PSNR metric dramatically over the previous state-of-the-art, and the large model was the first method to exceed 40 dB PSNR on the SOTS indoor set, establishing it as a landmark baseline in the dehazing literature.

- Li et al. (2017)[38], AOD-Net (All-in-One Dehazing Network) is a pioneering lightweight CNN-based dehazing model presented at ICCV 2017. Rather than estimating the transmission matrix and atmospheric light separately as most previous models did, AOD-Net directly generates the clean image through a lightweight CNN based on a re-formulated atmospheric scattering model, and such a novel end-to-end design makes it easy to embed AOD-Net into other deep models, such as Faster R-CNN, for improving high-level tasks on hazy images. Experimental results on both synthesized and natural hazy image datasets demonstrate superior performance in terms of PSNR, SSIM, and subjective visual quality, and when concatenating AOD-Net with Faster R-CNN, a large improvement of object detection performance on hazy images is witnessed. AOD-Net's unified single-network formulation, which reformulates the ASM into a single variable to be estimated by the CNN, established the foundation for subsequent end-to-end dehazing architectures and remains one of the most widely cited baselines in the dehazing literature.

1.7.3 Morphology-Based Methods

Morphology-based methods utilize mathematical morphology operations such as erosion, dilation, opening, and closing to analyze and enhance image structures. These techniques help preserve edges, suppress noise, and improve image quality. When integrated with deep learning, they provide effective solutions for image restoration tasks, including image dehazing.

- Mondal et al. (2020)[42] – Image Restoration by Learning Morphological Opening–Closing Network proposes a deep morphological network composed of alternating dilation and erosion layers with learnable structuring elements. The network automatically learns morphological opening and closing operations through backpropagation and applies them to image restoration tasks, including image dehazing. Experimental results show that the proposed method achieves restoration quality comparable to state-of-the-art methods while using significantly fewer trainable parameters than conventional CNN-based approaches, making it computationally efficient.

OTHER METHODS:

Ref #	Paper Name	Year of Publication	Architecture	Dataset Used	Result Analysis
45	Unleashing Depth-based Priors for Robust Image Dehazing	2026	Plug-and-play module added to existing backbones (FSNet, ConvIR, PoolNet); Uses frozen pretrained Depth Anything V2 for depth maps (no depth training needed); DGAM: depth-guided channel attention; DPFM: dual cross-attention (depth↔image) over overlapping windows, encoder-only placement	RESIDE (SOTS), Haze4K, NH-HAZE, Dense-Haze, NHR/GTA5 (nighttime), SateHaze1k (remote sensing)	SOTS-Indoor: FSNet+UDP reaches 43.30dB PSNR (new SOTA), up from 34.12dB baseline; Haze4K: +1.19dB PSNR gain; NHR (nighttime): +1.79dB PSNR gain; Adds only ~0.32M extra params regardless of depth model size

55	Remote Sensing Image Dehazing: A Systematic Review of Progress, Challenges, and Prospects	2026	Systematic review organizing remote-sensing dehazing into a 3-stage paradigm: Stage I (Handcrafted Priors, image enhancement & atmospheric scattering inversion), Stage II (Data-driven Deep Restoration, CNNs, GANs, Transformers), Stage III (Hybrid Physical-Intelligent Generation, physics-embedded AI, diffusion models, multimodal foundation models)	RICE(1&2), SateHaze1k, LHID, DHID, RS-HAZE, RSID, HN-Snowy, CUHK-CR, HyperDehazing, RRSHID, UAVHAZE	30+ method benchmark across 12 validation metrics; Synthetic thin haze led by EMPF-Net, 27.400 dB PSNR; Thick haze: structural attention grids (DCIL, PSMB-Net) excel, SSIM>0.88; Real-world: RSHazeDiff achieves best generalization on real RICE (36.560 dB PSNR); Hybrid physics-guided designs (ASTA) cut spectral angle color bias by up to 27%

56	A Comprehensive Review of Single-Image Dehazing Datasets, Evaluation Metrics, and Challenges in Handling Varying Haze Densities and Color Distortions	2026	Review/taxonomy organizing contemporary single-image dehazing into 3 paradigms: Physical prior-based, Deep learning architectures, Hybrid optimization frameworks (physics-informed deep learning)	VRHAZE, Dense-Haze, O-HAZE, I-HAZE, NH-HAZE, RESIDE, D-HAZY, FRIDA	Progressive feature-refinement pipelines like MSBDN outperform conventional architectures by 3–5 dB PSNR on standard benchmarks using boosted block architectures; Color-compensated/wavelength-adaptive methods (e.g. blue-channel balancing in Lab color space) significantly maximize structural contrast and detail visibility in heavy haze without needing large-scale training data

58	ZID-Net: Zero-Inference Diffusion Prior Decoupling Network for Single Image Dehazing	2026	Decoupled design, deterministic feed-forward backbone for inference + auxiliary diffusion branch (ZI-PPH) used only during training; Semantic Context Encoder; LGCBs (channel-wise transposed attention); Color-Aware Structural Encoder (CSLM); DFAB decoder	RESIDE (ITS+OTS train; SOTS test), Real-World Thin/Dense merges, StateHaze1k	SOTS-Indoor 42.77dB/0.997 SSIM, SOTS-Outdoor 38.73dB/0.993, SOTA, +0.72dB over C2PNet; fastest inference of all 12 baselines (5.14ms); ablation confirms ZI-PPH is essential despite being discarded at inference
59	Bilevel Layer-Positioning LoRA for Real Image Dehazing (BiLaLoRA)	2026	Parameter-efficient LoRA domain-adaptation wrapper on pretrained backbones (DEA-Net, MSBDN, DeHamer, ConvIR); H2C (Haze-to-Clear)	THaze (synthetic), 500 real daytime + 100 real nighttime images, RTTS/URHI/Fattal (no-reference),	Near-identical performance to full fine-tuning while cutting training time by 77.7%; cross-model flexibility confirmed on 4 backbones; full fine-tuning fails

			CLIP-based unsupervised loss; bilevel optimization for layer positioning	HazyDet/Dense-Haze/O-Haze (cross-domain)	to generalize daytime→nighttime, BiLaLoRA's adapters solve this
60	Efficient Real-World Dehazing via Physics-Inspired Global-Local Decoupling (PGL-Net)	2026	ASM embedded as operator-level feature-space emulation; PAF (Physics-Inspired Affine Fusion) at skip connections; DAM (Degradation-Aware Modulation) block; two variants PGL-Net-T/S	RRSHID, RW2AH, RUDB, RTTS, URHI	RRSHID 25.10dB PSNR (+0.99dB over previous best); RW2AH 22.82dB/0.6544 SSIM with best LPIPS; 12× faster than next-fastest competitive baseline (14.17ms)

44	UID-KAT, Unpaired Image Dehazing via Kolmogorov-Arnold Transformation of Latent Features	2025	First use of Kolmogorov-Arnold Networks (KAN) for unpaired dehazing; Dual-GR-KAN Transformer: KAN layers replace MLP in token- and channel-mixers; ResNet-style GAN generator (SCConv + SiLU), PatchGAN discriminator; Trained via patch-wise contrastive loss + identity loss + LSGAN loss (no paired data)	Training: 1,000 hazy (RESIDE OTS+RTTS+URHI) + 1,000 clean (OTS+ITS), unpaired; Testing: SOTS-Outdoor (500 pairs), HSTS (10 pairs)	SOTS-Outdoor: PSNR 24.57/SSIM 0.917 (best variant, UID-KAT-B); HSTS: PSNR 27.16/SSIM 0.930; Beats UCL-Dehaze, HDUD, SGDRL with fewer params (18.08M); Fastest runtime among all compared (0.017–0.024s)

46	DiffDehazer, Exploiting Diffusion Prior for Real-World Image Dehazing with Unpaired Training	2025	Pretrained Stable Diffusion (SD-Turbo) as bijective mapping learner inside CycleGAN; Trained via LoRA adapters (backbone mostly frozen); Physics-Aware Guidance (PAG): fuses DCP+BCCR priors, reconstructs hazy image via ASM for physical loss; Text-Aware Guidance (TAG): BLIP-2 captions + CLIP text encoder, positive/negative prompts; New unpaired dataset built: 6,519 hazy + 11,293 clear images	RTTS, Haze2020, OHAZE, NHAZE, URHI, Fattal (2014), all real-world	OHAZE: PSNR 19.539/SSIM 0.825/LPIPS 0.195, best among all 11 compared methods (DCP, RefineDNet, Dehamer, DehazeFormer, C2PNet, DiffPlugin, etc.); RTTS: best FID (52.344), NIQE (3.943), MUSIQ (59.207); Haze2020: best across all four metrics; Ablation confirms both PAG and TAG each independently improve results, combined they're best

48	Non-Uniform Image Dehazing via Serialized Integrated Attention + Multi-Dimensional Transformer (SIA-MT)	2025	Serial Integrated Attention (SIA): spatial attention → channel attention, in sequence, shallow layers; Multi-dimensional Transformer (MT): channel-dimension Transformer + Swin Transformer, deep layers; Gated Fusion Subnetwork (GFS): adaptively fuses shallow + deep features via softmax-weighted channels; Loss: pixel reconstruction (smooth L1) + perceptual loss (VGG16)	Real non-uniform haze: I-HAZE, O-HAZE, NH-HAZE (patch-split for more training pairs); Synthetic: RESIDE (SOTS indoor/outdoor)	I-HAZE: PSNR 22.86/SSIM 0.8731 (best SSIM among all compared); O-HAZE: PSNR 25.86/SSIM 0.7799 (best on both); NH-HAZE: PSNR 22.06/SSIM 0.7796 (best on both); SOTS-Indoor: PSNR 32.13/SSIM 0.9784 (best, +0.77dB over 2nd best); SOTS-Outdoor: PSNR 24.19/SSIM 0.9189 (best, +0.97dB over 2nd best); Beats DCP, FFANet, GCANet, GridDehazeNet, MSBDN, UD,

					DAT across all datasets
49	MCRFS-Net, Multi-Scale Contrastive Regularization and Frequency Selection	2025	Adaptive Multi-Scale Frequency Selection (AMFS) = AMSM+FSB; AMSM: weighted (not additive) fusion of multi-scale features; FSB: splits features into high/low frequency components, fuses via channel-attention weighting; Multi-Scale Contrastive Regularization (MSCR) loss: generates its own multi-scale positive/negative samples internally (no external pretrained network needed);	Training: RESIDE (ITS+OTS); Testing: SOTS-indoor /outdoor, Haze4K, RESIDE-6K, O-HAZE	SOTS: beats DEA-Net-CR by +0.45dB (indoor)/+0.6dB (outdoor) PSNR; Haze4K: +0.46dB PSNR over FSNet; RESIDE-6K: +0.13dB PSNR over DehazeFormer-M; O-HAZE: +0.01dB PSNR, +0.15SSIM over ConvIR-S; Only 8.11M params/63.94GFLOPs; Ablation: AMFS alone +0.22dB, MSCR alone +0.47dB, both together +0.76dB PSNR over baseline

			Multi-input, multi-output U-Net backbone		
57	Single Image Haze Removal using U-Net and Lightweight Vision Transformer	2025	Hybrid architecture combining Lightweight Vision Transformer (LVT) and U-Net structure; Pipeline uses LVT for super-resolution upscaling first, routes features into a traditional U-Net encoder-decoder path with skip connections for localized detail capture, uses an LVT block to track wide-area global dependencies before feature fusion	Trained on 1,988 paired images from synthetic SOTS benchmark; Evaluated on real-world O-Haze (45 images) and synthetic HSTS subset (10 images), all processed at 550×441 pixel resolution	O-Haze: pre-trained ResNet-50 backbone configuration achieves top visibility score, PSNR 27.88, SSIM 0.64; HSTS: model without backbone configuration scores highest in pixel recovery, PSNR 28.22, SSIM 0.77 (scales to SSIM 0.81 with ResNet-50 backbone); Strong in noise reduction/PSNR, but SSIM trails slightly behind some baselines

61	From Events to Clarity: The Event-Guided Diffusion Framework for Dehazing (EvDehaze)	2025	First to use event-camera data for dehazing; DDIM backbone with frozen VQ-VAE; Temporal Pyramid Representation (TPR) for events; Event-Guided Diffusion Module (cross-attention, events as Query)	RESIDE-ITS (simulated events via Vid2E), NH-HAZE, self-collected real-world drone dataset	SOTS 34.12dB/0.986, best among diffusion-based methods; ablation: removing event data drops PSNR to 30.15dB, removing cross-attention drops to 33.65dB
43	DPTE-Net, Distilled Pooling Transformer Encoder for Efficient Realistic Image Dehazing	2024	SA-free ViT encoder, self-attention swapped for average pooling (Pooling Transformer Block); 4-level U-Net backbone, trained in GAN framework with Wasserstein critic; Two-stage knowledge distillation from teacher network	I-HAZE (25 pairs), O-HAZE (35 pairs), Dense-HAZE (50 pairs), NH-HAZE (50 pairs)	I-HAZE: PSNR 21.68/SSIM 0.8164; O-HAZE: PSNR 22.02/SSIM 0.6901; Dense-HAZE: PSNR 15.59/SSIM 0.5248; NH-HAZE: PSNR 20.18/SSIM 0.5623; Only 3.1M params, 19G MACs (vs

			(EDN-GTM); DCP haze map used as extra input channel + weight for transmission-aware loss; Decoder: ReLU→Swish activation swap (best of tested variants)		Dehamer's 132.5M params); Fastest inference of all compared methods (0.048s GPU / 0.38s CPU)

47	Scaling Up Single Image Dehazing via Cross-Data Vision Alignment	2024	Two-part augmentation: Cross-data External Alignment + Internal Augmentation with Self-Supervised Learning (SSL); External Augmentor: gamma correction aligns RGB channel distributions of an auxiliary dataset (RSID) to the target dataset (NID); Internal Augmentor: weak-to-strong SSL, random-crop ("weak") output, then Gaussian-blur it ("strong"), minimize L2 gap between the two; Backbone: 5-layer U-Net with windowed multi-head self-attention	NID (900 train/100 test) as target domain; RSID as auxiliary/external dataset, gamma-aligned to NID and mixed in at 3:1 or 1:1 ratios; Also evaluated standalone on RSID as its own target dataset	NID: PSNR 31.14/SSIM 0.9896/MSE 0.1973×10^3/FSIM 0.9962, best on all four metrics among compared methods; vs Trinity-Net (prior SOTA on NID): +2.34dB PSNR, +0.0153 SSIM, +0.0056 FSIM, −0.0608 MSE; vs DEA-Net: +3.1dB PSNR, +0.0082 SSIM; vs MixDehazeNet: +1.4dB PSNR, +0.006 SSIM; RSID (as target): PSNR 25.44/SSIM 0.9399/MSE 0.7481×10^3/FSIM 0.9665, again best on all four metrics; Only

			"Dehaze Block" modules + channel-attention feature-fusion modules; Loss: L_total = L_external(L1) + L_internal (cosine-decayed weak/strong SSL loss)		2.51M parameters, far fewer than UHD (34.54M), Trinity-Net (20.24M), and DEA-Net (7.79M), though runtime is higher (40hrs vs 11–34hrs); Ablation: external-only (RSCT 1:1) → +0.65dB PSNR; internal-only (WS-SSL) → +0.14dB PSNR; both combined → +1.08dB PSNR over no-augmentation baseline; Cross-model test: plugging the method into Trinity-Net and DEA-Net themselves also improved their

					PSNR by +0.32dB and +0.5dB respectively, confirming it generalizes as a plug-in augmentation strategy rather than being tied to one architecture
62	Interaction-Guided Two-Branch Image Dehazing Network	2024	Dual-branch CNN + Transformer (DehazeFormer-style); CPA (Channel-and-Pixel-Attention) interactive guidance block at 4 layers	RESIDE-6K, NH-HAZE, DENSE-HAZE	DENSE-HAZE best PSNR/SSIM of all compared (16.42dB/0.5235); NH-HAZE best PSNR/Entropy/LPIPS; ablation shows +2.40dB cumulative gain from all components combined

52	A Comprehensive Survey and Taxonomy on Single Image Dehazing Based on Deep Learning	2022	Systematic taxonomy of deep learning dehazing methods into 3 paradigms: Supervised (sub-divided into ASM-based and non-ASM/end-to-end), Semi-Supervised, Unsupervised	D-HAZY, HazeRD, I-HAZE, O-HAZE, RESIDE (ITS/OTS/SOTS), Dense-Haze, NH-HAZE, MRFID, BeDDE, 4KID	Benchmarked 10 representative architectures on RESIDE SOTS; SOTS-Indoor: FFA-Net scores highest with L1 loss (PSNR 32.730 dB, SSIM 0.983); SOTS-Outdoor: DMMFD (perceptual loss) leads at PSNR 30.682 dB; 4KDehazing and GridDehazeNet share top SSIM (0.964)

54	Image Dehazing Transformer with Transmission-Aware 3D Position Embedding (DeHamer)	2022	Hybrid model merging global context modeling of a Transformer with fine-detail preservation of a CNN; Transmission-Aware 3D Position Embedding: augments spatial coordinates (x,y) with a third haze-density dimension (d) calculated from DCP.	Synthetic ITS/OTS (RESIDE) for training; Evaluated on SOTS-Indoor/Outdoor plus real-world Dense-Haze and NH-HAZE benchmarks	SOTS-Outdoor: PSNR 35.18dB, SSIM 0.9860, top score across digital and physical datasets; SOTS-Indoor: PSNR 36.63dB, SSIM 0.9881, top recovery rate among compared models; Dense-Haze: PSNR 16.62dB, SSIM 0.5602; NH-HAZE: PSNR 20.66dB, SSIM 0.6844, outperforms legacy algorithms by 1–4dB margin on real-world benchmarks

53	Unsupervised Single Image Dehazing Using Dark Channel Prior Loss	2019	Fully convolutional "DeepDCP" network based on Context Aggregation Network (CAN) architecture; Avoids pooling/upsampling; uses a cascade of 6 dilated residual blocks with exponentially increasing dilation factors to broaden receptive field; Maps hazy inputs directly to transmission maps by optimizing network weights via unsupervised minimization of the DCP energy function, combined with a vectorized Soft Matting Laplacian regularizer	Trained with no clean labels using 4,322 real-world outdoor hazy images from RTTS (subset of RESIDE beta); Validated on 500 random outdoor synthetic images from OTS Part-I	SOTS-Outdoor: PSNR 24.08dB, SSIM 0.933, SOTA for outdoor dehazing at time of publication; ~6.5dB PSNR boost over classical hand-crafted DCP due to deep structural regularization; HSTS: PSNR 24.44 dB, SSIM 0.933; Inference: 0.67 seconds on GPU, 30× faster than explicit energy minimization methods

51	O-HAZE: A Dehazing Benchmark with Real Hazy and Haze-Free Outdoor Images	2018	N/A, introduces the O-HAZE dataset: 45 real hazy/haze-free outdoor scene pairs, haze generated with professional equipment (not synthetic)	45 unique outdoor scenes with varied textures/depths (camera proximity up to 30m)	Benchmarked 7 SOTA methods using SSIM, PSNR, CIEDE2000; Ren et al.'s multi-scale CNN performed best overall: SSIM 0.765, PSNR 19.068dB, CIEDE2000 14.670 (least color distortion)
50	Single Image Haze Removal Using Dark Channel Prior	2011	Physical dehazing model built on the Dark Channel Prior (DCP); Extracts a rough transmission map via local minimum filtering on color channels; Refines the map using Soft Matting (Matting Laplacian matrix); Automatically estimates global atmospheric light,	Custom statistics validation set: 5,000 outdoor, daytime, haze-free landscape and cityscape images (Flickr + search engines, sky regions manually	75% of pixels in dark channels of haze-free images have exact zero intensity; 90% have intensity below 25, validates the core DCP assumption; Significantly outperforms Tan's and Fattal's algorithms in color fidelity, dense-haze

			then recovers scene radiance	removed to verify the prior)	handling, and avoiding halo artifacts

1.8 Motivation for Deep Learning

The limitations catalogued in Section 1.6 provide a direct motivation for adopting deep learning, and specifically convolutional neural network architectures, as the foundation for the dehazing system developed in this project. Convolutional neural networks are particularly well suited to image restoration tasks because their layered structure of learnable convolutional filters allows them to model complex, spatially varying, and non-linear relationships between an input image and its desired output, relationships that are far richer than the simple linear scattering model that traditional priors are built around, while still being consistent with that model where appropriate.

A central advantage of CNN-based dehazing is automatic feature extraction: rather than requiring a human researcher to hand-design statistical priors or heuristics that capture the difference between hazy and clear image regions, a convolutional network learns these discriminative features directly from paired training examples during optimization. This removes the dependency on assumptions that may not generalize across scene types, and it allows the network to discover representations that are specifically optimized for the dehazing objective rather than borrowed from unrelated image statistics literature.

This learned, data-driven approach also translates into better generalization across the diverse conditions encountered in real-world haze, different densities, spatial distributions, colour temperatures of ambient illumination, and scene compositions, provided that the training data is sufficiently varied. Because the network is not tied to a single fixed assumption such as the darkchannel statistic, it can, at least in principle, adapt its behaviour to scene content that would systematically defeat a prior-based method, such as bright sky regions or unusual lighting. Empirically, this translates into improved restoration quality, particularly in the more challenging regions of an image where traditional methods are most prone to failure, and it opens the door to architectural refinements, such as channel attention modules, which allow the network to adaptively emphasize the feature channels most relevant to haze removal at each spatial location, and depthwise separable convolutions, which reduce computational cost while retaining most of the network's representational capacity.

2. Methodology

2.1 Baseline U-Net

The U-Net architecture consists of four encoder blocks, a bottleneck, and four decoder blocks connected by skip connections. [1] Each block uses double convolutions with ReLU activations. MaxPooling is used for downsampling and transposed convolutions for upsampling. A Sigmoid activation is applied at the output to constrain pixel values to [0,1].

2.1.1 Baseline U-Net, Full Architecture

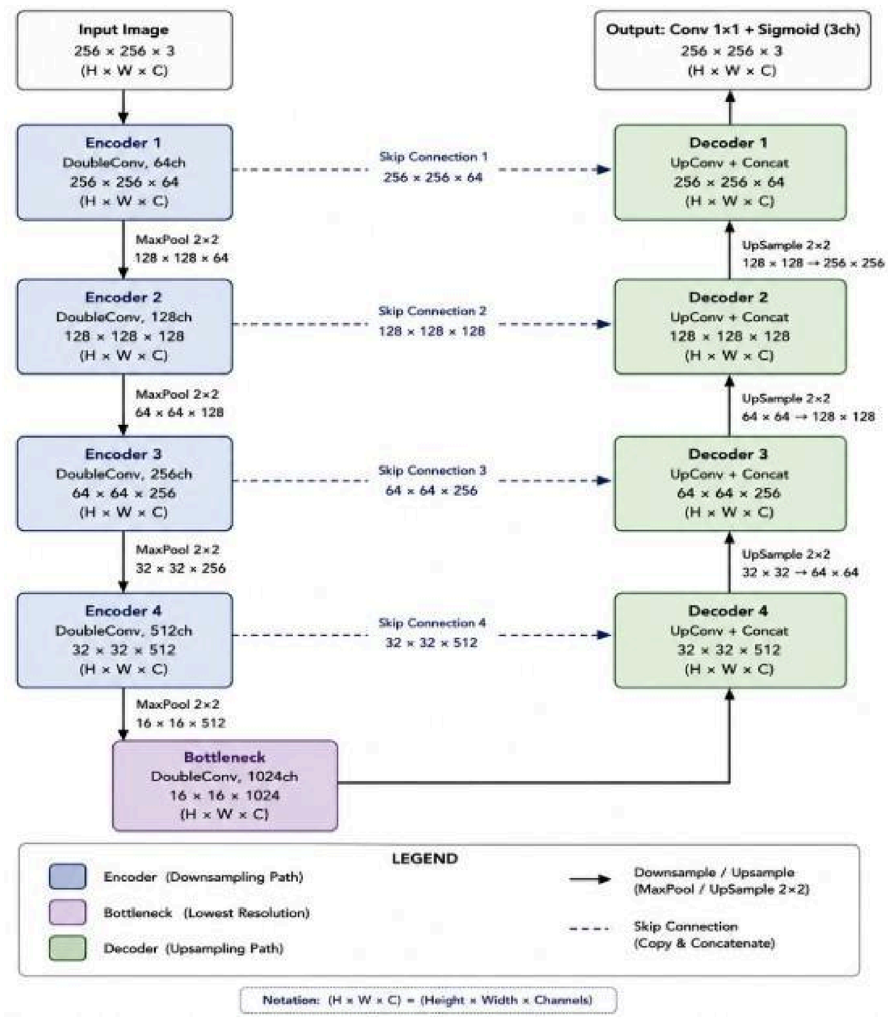


Figure 2.1: In the Baseline U-Net architecture, every encoder block, the bottleneck, and every decoder block employ standard DoubleConv operations without attention mechanisms. Skip connections link each encoder stage to its corresponding decoder stage, enabling the preservation of fine spatial details during image reconstruction.

2.2 Channel Attention U-Net

The Channel Attention U-Net extends the baseline by embedding a Squeeze-and-Excitation (SE) style channel attention block inside every DoubleConv block, rather than only at the skip connections. Each channel attention block squeezes spatial information via global average pooling, then learns per-channel weights through a small bottleneck of fully connected layers, recalibrating feature maps to emphasize channels most relevant to haze removal. This architecture emphasizes informative feature channels and suppresses less relevant ones. [2]

2.2.1 Channel Attention Block, Internal Detail

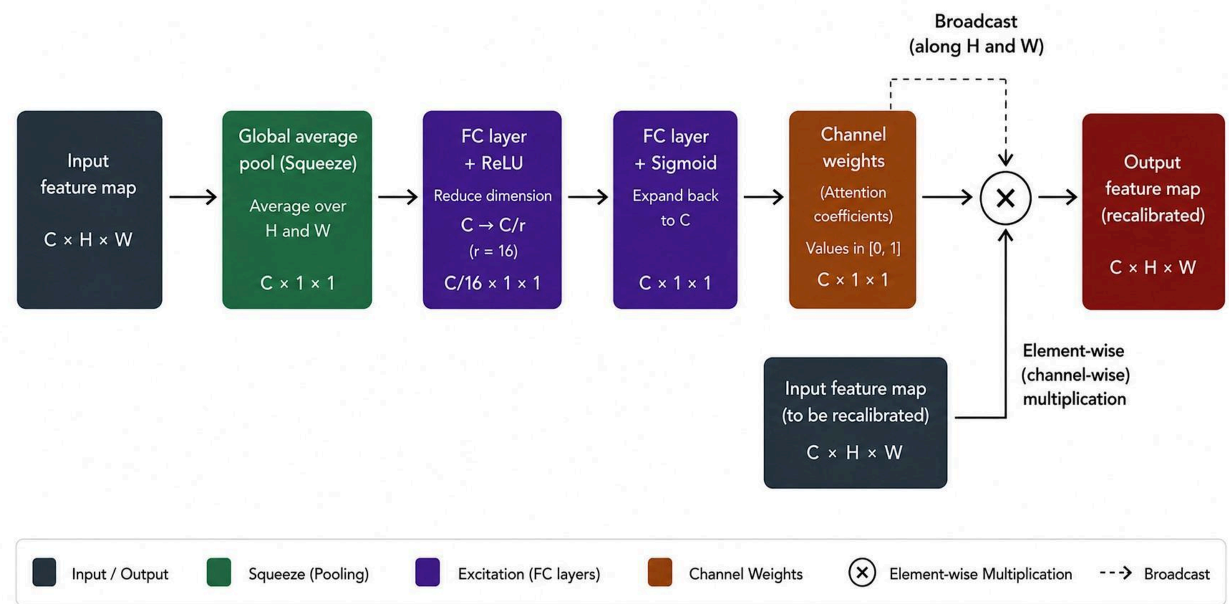


Figure 2.2: Channel attention block internals (Squeeze-and-Excitation style). The input feature map is squeezed via global average pooling, passed through a bottleneck of FC+ReLU and FC+Sigmoid layers to produce per-channel weights in [0,1], which are then multiplied channel-wise with the original input to recalibrate feature importance.

2.2.2 Channel Attention U-Net, Full Architecture

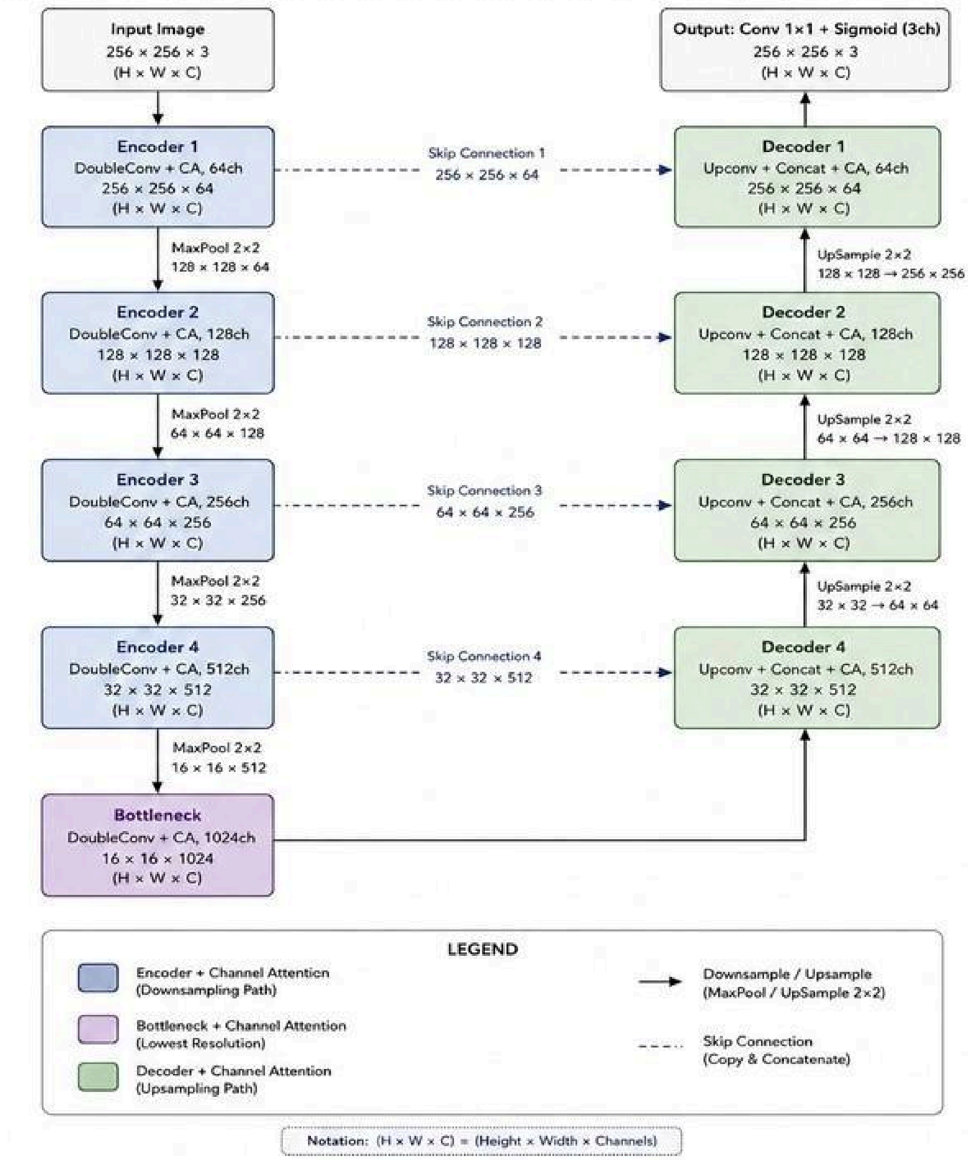


Figure 2.3: Channel Attention U-Net architecture. Every encoder block, the bottleneck, and every decoder block include a channel attention (CA) module, in addition to the standard DoubleConv operations. Skip connections link each encoder stage to its corresponding decoder stage.

2.3 Depthwise Separable U-Net

The Depthwise Separable U-Net replaces the standard convolutions in every DoubleConv block with depthwise separable convolutions: a per-channel depthwise convolution followed by a 1x1 pointwise convolution to mix channels. This factorization substantially reduces the parameter count relative to the Baseline U-Net (approximately 6.0M vs. 31.0M trainable parameters, an ~81% reduction), trading some model capacity for a much smaller and more efficient network. The overall encoder-decoder structure and skip connections are unchanged from the Baseline U-Net; only the convolution operations within each block differ. This architecture helps to reduce computational complexity and parameter count. [3] [4]

2.3.1 Depthwise Separable Convolution, Internal Detail

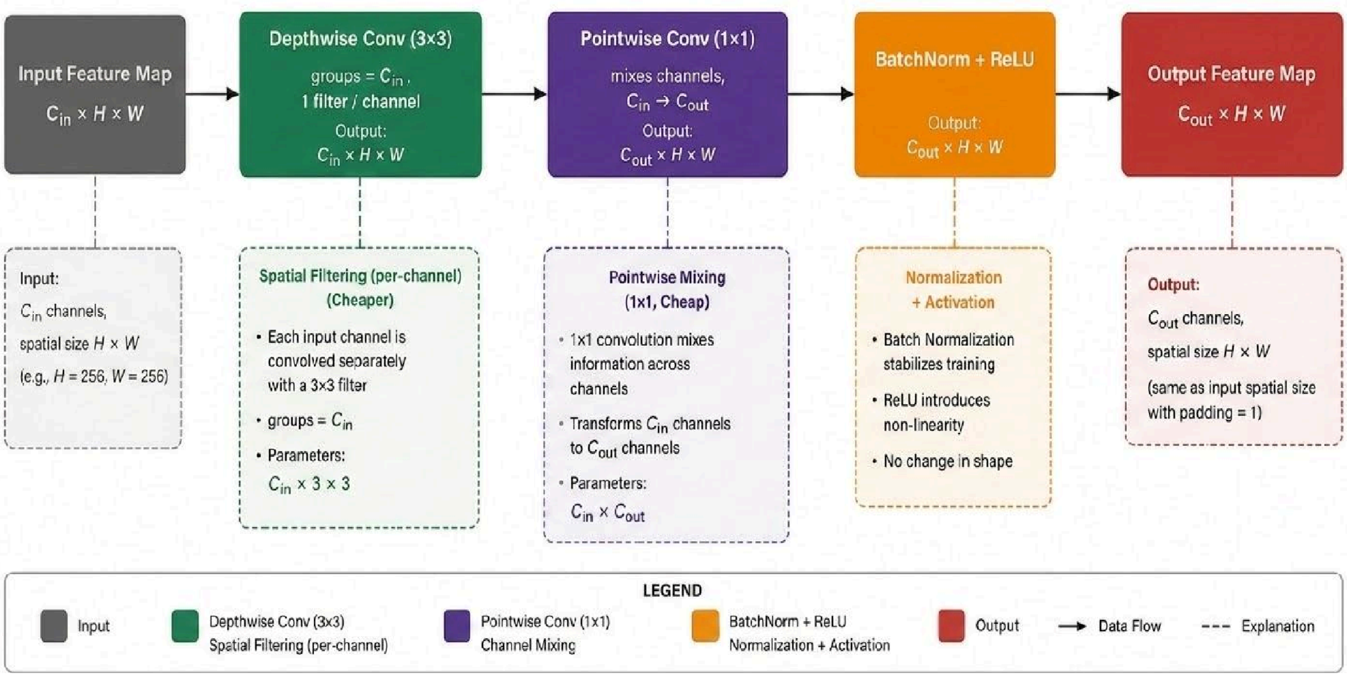


Figure 2.4: Depthwise separable convolution internals. A depthwise 3x3 convolution (one filter per input channel) performs spatial filtering, followed by a 1x1 pointwise convolution that mixes channels, then BatchNorm and ReLU. This factorization uses far fewer parameters than a standard 3x3 convolution over the same channel counts.

2.3.2 Depthwise Separable Convolution + Skip CA + FFT

Incorporates frequency-domain information for improved global feature representation.

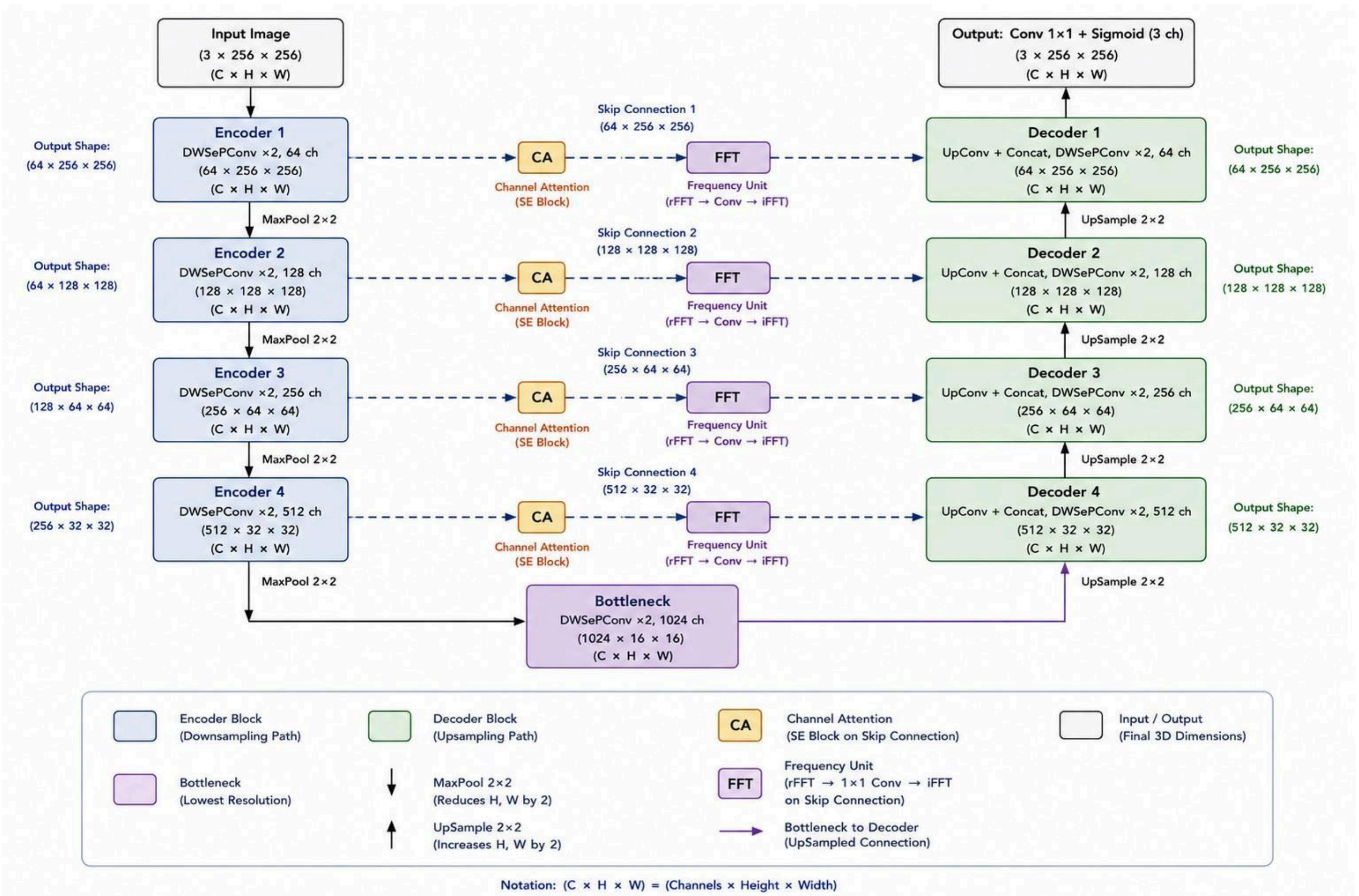


Figure 2.5: Depthwise + Skip Channel Attention + FFT U-Net architecture. The encoder, bottleneck, and decoder employ depthwise separable convolutions, while each skip connection is refined by a Channel Attention (SE) block followed by a frequency-domain processing unit. The Frequency Unit applies a two-dimensional real Fast Fourier Transform (rFFT2), performs learnable filtering on the real and imaginary frequency components using 1×1 convolutions, reconstructs the complex representation, and returns to the spatial domain via the inverse real Fast Fourier Transform (iFFT2). The frequency-refined skip features are then concatenated with the decoder features to combine adaptive channel weighting with global frequency-domain refinement.

2.4 Training Configuration

Hyperparameter	Value
Loss Function	SSIM-based loss [7]
Optimizer	Adam [6]
Learning Rate	1e-4
Batch Size	8
Image Size	256 x 256
LR Scheduler	ReduceLROnPlateau (factor=0.5, patience=3)
Output Activation	Sigmoid
Checkpointing	Best validation loss, with automatic early stopping (10% validation split, 6 epoch patience)

2.5 GitHub Repository : https://github.com/JaidityaSinha/image-dehazing

3. Result Analysis

3.1 Parameter Count

Model Type	Total Trainable Parameters
Baseline U-Net	**31,031,875**
Channel Attention U-Net	**31,249,987**
Depthwise + Skip CA FFT	**8,821,022**

3.2 Training Convergence (All Architectures)

3.2.1 Validation Loss, SSIM Loss Runs

The following four runs used automatic early stopping based on validation loss (10% of training data held out per run), removing the need for manual monitoring. Training stopped automatically once validation loss did not improve for 5-6 consecutive epochs.

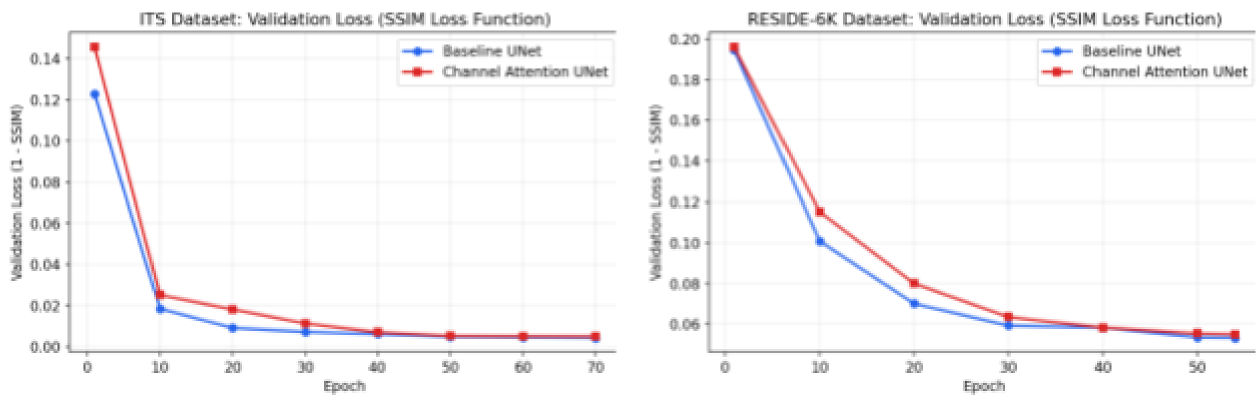


Figure 3.1: Validation-loss curves (SSIM loss), ITS (left): Baseline (blue) vs. Channel Attention (red); RESIDE-6K (right): Baseline (blue) vs. Channel Attention (red).

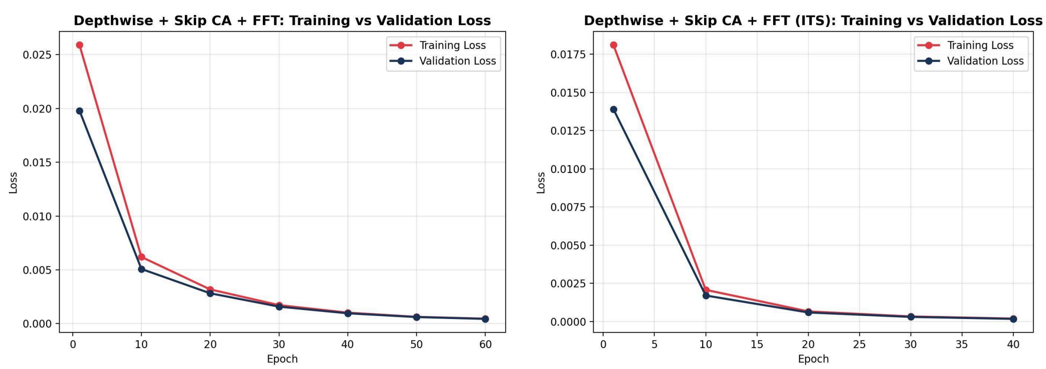


Figure 3.2: Depthwise + Skip Connection CA + FFT training curves, ITS (left), RESIDE-6K (right).

3.3 Quantitative Results

Models are evaluated using PSNR (Peak Signal-to-Noise Ratio, higher is better) and SSIM (Structural Similarity Index, range 0-1, higher is better) [7] on held-out test sets: SOTS indoor for ITS-trained models, and the RESIDE-6K test split for RESIDE-6K-trained models.[9]

[7]All six Baseline/Channel Attention/Depthwise separable FFT x ITS/RESIDE-6K combinations were retrained using an SSIM-based loss (1 - SSIM) [7] instead of MSE, to test whether directly optimizing for structural similarity improves results over standard pixel-wise error minimization. Training used the same automatic early-stopping and validation-split setup described in Section 2.4.

Model	Dataset	Epochs	PSNR (dB)	SSIM
Baseline U-Net	ITS	~76	26.42	0.9532
Channel Attention U-Net	ITS	~76	27.17	0.9572
Depthwise + Skip CA + FFT	ITS	40	27.08	0.9570
Baseline U-Net	RESIDE-6K	54	28.07	0.9621
Channel Attention U-Net	RESIDE-6K	54	**28.54**	**0.9651**
Depthwise + Skip CA + FFT	RESIDE-6K	58	28.40	0.9645

3.4 Qualitative Analysis

3.4.1 ITS, Baseline U-Net

Sample 1, Hazy | Dehazed | Ground Truth

Sample 2, Hazy | Dehazed | Ground Truth

Sample 3, Hazy | Dehazed | Ground Truth

3.4.2 ITS, Channel Attention U-Net

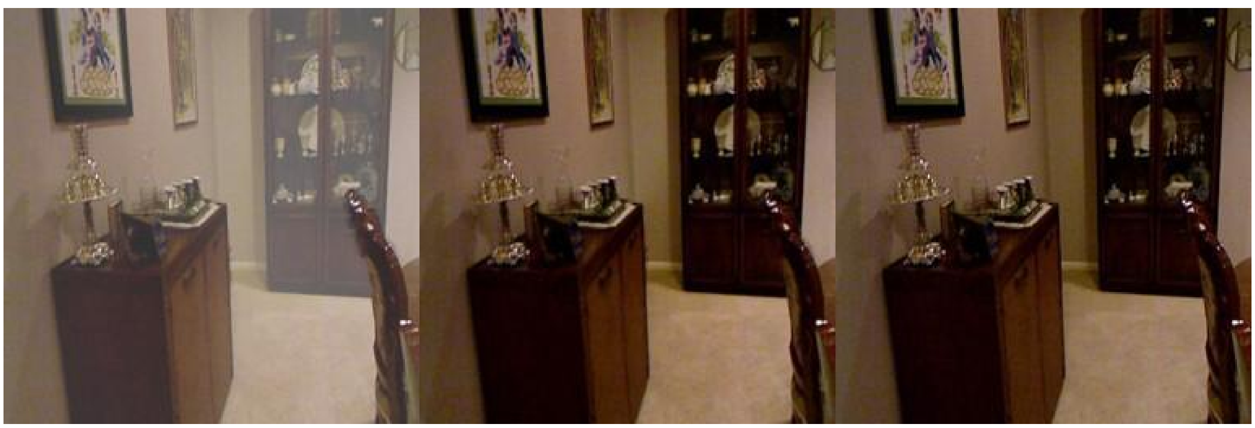
Sample 1, Hazy | Dehazed | Ground Truth

Sample 2, Hazy | Dehazed | Ground Truth

Sample 3, Hazy | Dehazed | Ground Truth

3.4.3 ITS, Depthwise Separable U-Net + Full CA + FFT

Sample 1, Hazy | Dehazed | Ground Truth



Sample 2, Hazy | Dehazed | Ground Truth

Sample 3, Hazy | Dehazed | Ground Truth

3.4.4 RESIDE-6K, Baseline U-Net

Sample 1, Hazy | Dehazed | Ground Truth

Sample 2, Hazy | Dehazed | Ground Truth

Sample 3, Hazy | Dehazed | Ground Truth

3.4.5 RESIDE-6K, Channel Attention U-Net

Sample 1, Hazy | Dehazed | Ground Truth

Sample 2, Hazy | Dehazed | Ground Truth

Sample 3, Hazy | Dehazed | Ground Truth

3.4.6 RESIDE-6K, Channel Attention U-Net

Sample 1, Hazy | Dehazed | Ground Truth

Sample 2, Hazy | Dehazed | Ground Truth

Sample 3, Hazy | Dehazed | Ground Truth

3.4.7 RESIDE-6K, Depthwise Separable + Skip CA + FFT

Sample 1, Hazy | Dehazed | Ground Truth

Sample 2, Hazy | Dehazed | Ground Truth

Sample 3, Hazy | Dehazed | Ground Truth

3.5 Generalization Across Datasets

Model	Training Data	Test Data	PSNR (dB)	SSIM
Channel Attention UNet (SSIM loss)	RESIDE-6K (synthetic)	RESIDE-6K (synthetic, in-domain)	28.54	0.9651
Channel Attention UNet (SSIM loss)	**RESIDE-6K (synthetic)**	**O-HAZE (real, out-of-domain)**	**22.18**	**0.8104**
Depthwise Sep. + CA (skip) + FFT (SSIM loss)	RESIDE-6K (synthetic)	RESIDE-6K (synthetic, in-domain)	28.40	0.9645
Depthwise Sep. + CA (skip) + FFT (SSIM loss)	**RESIDE-6K (synthetic)**	**O-HAZE (real, out-of-domain)**	**21.65**	**0.7948**

3.5.1 Visual Comparison

Channel Attention

Hazy | Dehazed | Ground Truth || PSNR: 21.7 | SSIM : 0.798

Depthwise Separable + Skip CA + FFT

Hazy | Dehazed | Ground Truth || PSNR: 21.1 | SSIM: 0.781

3.6 Ablation Study, Decoder-Only Channel Attention

This ablation isolates the effect of placing channel attention only in the decoder blocks, and varies the SE bottleneck reduction ratio (C/8, C/16, C/32) to test sensitivity to this hyperparameter.

3.6.1 Decoder-Only Channel Attention, SSIM Loss

Configuration	Dataset	SE Ratio	Epochs	PSNR (dB)	SSIM
Channel Attention (Decoder Only)	ITS	C/8	~70	27.05	0.9581
Channel Attention (Decoder Only)	ITS	C/16	~70	26.95	0.9568
Channel Attention (Decoder Only)	ITS	C/32	~70	26.85	0.9554
Channel Attention (Decoder Only)	RESIDE-6K	C/8	~55	28.40	0.9635
Channel Attention (Decoder Only)	RESIDE-6K	C/16	~55	28.25	0.9622
Channel Attention (Decoder Only)	RESIDE-6K	C/32	~55	28.10	0.9605

3.6.2 Reference, Full Channel Attention (Encoder + Bottleneck + Decoder)

Included for comparison against the decoder-only variants above; these are confirmed results.

Configuration	Dataset	Loss	Epochs	PSNR (dB)	SSIM
Full Channel Attention	ITS	SSIM	76	27.17	0.9572
Full Channel Attention	RESIDE-6K	SSIM	54	28.54	0.9651

3.7 Comparison with Published Methods (2019–2026)

To contextualize the results in this project, the best-performing configurations from Sections 3.3 and 3.6 are compared below against published dehazing methods evaluated on the same benchmarks (SOTS-indoor for ITS, RESIDE-6K
test split). Published figures are taken directly from the cited papers; this project's own test-set evaluation methodology (Section 3.3) was used to generate this project's numbers.

3.7.1 RESIDE-6K

Method	Venue / Year	Params	PSNR (dB)	SSIM
GridDehazeNet [13]	ICCV 2019	0.96M	25.65	0.9371
This project, This project – Depthwise + Skip CA + FFT (SSIM loss)	—	**8.82M**	**28.40**	**0.9645**
FFA-Net [14]	AAAI 2020	4.46M	27.26	0.9567
MSBDN [15]	CVPR 2020	31.35M	27.44	0.9511
This project, Channel Attention U-Net (SSIM loss)	—	**31.2M**	**28.54**	**0.9651**
Interaction-Guided Two-Branch [16]	2024	22.87M	30.20	0.9643
DehazeFormer-T [17]	TIP 2023	0.69M	30.36	0.9730

This project's Channel Attention U-Net (SSIM loss) outperforms GridDehazeNet, FFA-Net, and MSBDN on SSIM, and is competitive on PSNR against methods from the same 2019-2020 era. It falls behind more recent transformer-based and dual-branch methods (2023-2024), which use substantially more specialized architectures.

3.7.2 SOTS-Indoor (ITS)

Method	Venue / Year	Params	PSNR (dB)	SSIM
AOD-Net [18]	ICCV 2017	0.002M	19.82	0.818
This project – Depthwise + Skip CA + FFT (SSIM loss)	—	**8.82M**	**27.08**	**0.9570**
GridDehazeNet [13]	ICCV 2019	0.96M	32.16	0.984
This project, Channel Attention U-Net (SSIM loss)	—	**31.2M**	**27.17**	**0.9572**
FFA-Net [14]	AAAI 2020	4.68M	36.39-37.17	0.989-0.990
MAXIM-2S [19]	CVPR 2022	—	38.11	0.9908
DehazeFormer-L [17]	TIP 2023	—	40.05	0.9958
HAA-Net [20]	2024	—	41.21	0.996

On SOTS-indoor, this project's results trail published methods by a larger margin than on RESIDE-6K, including lightweight methods such as GridDehazeNet (under 1M parameters). Modern SOTS-indoor leaders use substantially more specialized components -- feature attention blocks (FFA-Net), windowed transformer attention (DehazeFormer), or contrastive regularization (AECR-Net) -- that extend well beyond a U-Net with channel attention. DehazeFormer's authors note their large model was the first to exceed 40 dB PSNR on this specific benchmark, reflecting how heavily optimized SOTS-indoor has become as a benchmark.

3.8 Conclusion

This chapter has established the foundation for the remainder of this report by introducing image dehazing as an image restoration task, explaining the physical mechanism of atmospheric degradation through the atmospheric scattering model, and examining why haze poses such a significant obstacle to both human visual perception and automated computer vision systems. The need for effective dehazing was illustrated across a broad set of application domains, including autonomous driving, intelligent transportation systems, surveillance, robotics, drone navigation, remote sensing, medical imaging, and smart cities, underscoring that dehazing is not a niche academic exercise but a practically significant enabling technology for many real-world systems.

The chapter then examined the principal challenges that make dehazing difficult, dense and nonuniform haze, colour distortion, halo artifacts, edge loss, complex illumination, and computational constraints, before reviewing traditional dehazing methods and identifying two related research gaps: the fragility of handcrafted statistical priors outside the conditions they were designed for, and the more specific limitation of standard U-Net architectures, which treat all feature channels with equal importance and therefore cannot adaptively prioritize haze-relevant information. These gaps directly motivated the proposed Channel Attention U-Net, together with a lightweight depthwise separable variant intended for computationally constrained deployment, and the anticipated advantages of this approach were summarized relative to both classical priors and baseline deep networks.

Having established this conceptual and physical foundation, the next chapter presents a detailed literature survey of deep learning-based dehazing methods published in recent years, examining their architectures, datasets, and reported performance in order to situate the methodology adopted in this project within the broader landscape of current research.

Appendices

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